

2020 AMC 10B Problem 19 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 19. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 19

Problem:

In a certain card game, a player is dealt a hand of 10 cards from a deck of 52 distinct cards. The number of distinct (unordered) hands that can be dealt to the player can be written as $158A00A4A40$. What is the digit A ?

Answer Choices:

- A. 2
- B. 3
- C. 4
- D. 6
- E. 7

Solution:

We're looking for the number of ways we can get 10 cards from a deck of 52, which is represented by $\binom{52}{10}$.

$$\binom{52}{10} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44 \cdot 43}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

And after simplifying, we get

$$26 \cdot 17 \cdot 7 \cdot 47 \cdot 46 \cdot 5 \cdot 11 \cdot 43$$

Now, if we examine the number

$$158A00A4A40$$

we can notice that it is equal to some number n times 10 . Therefore, we can divide 10 from the aforementioned expression and find the unit digit, which will be A .

Now, after dividing ten, we will have

$$26 \cdot 17 \cdot 7 \cdot 47 \cdot 23 \cdot 11 \cdot 43$$

We can then use modulo 10 and find that the unit digit of the expression is $(A) \boxed{2}$.

OR

$$\binom{52}{10} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44 \cdot 43}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 5 \cdot 7 \cdot 47 \cdot 46 \cdot 11 \cdot 43$$

Since this number is divisible by 4 but not 8 , the last 2 digits must be divisible by 4 but the last 3 digits cannot be divisible by 8 . This narrows the options down to 2 and 6 .

Also, the number cannot be divisible by 3 . Adding up the digits, we get $18 + 4A$. If $A = 6$, then the expression equals 42 , a multiple of 3 . This would mean that the entire number would be divisible by 3 , which is not what we want

Therefore, the only option is $(A) \boxed{2}$.

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